

Physics 0525: Lab 1

Voltage and Current Dividers - measuring DC voltage and current

Equipment: Breadboard, digital multimeter (DMM), resistors.

Preliminaries: Read the description of the breadboard and the digital multimeter carefully. It is important to understand how they work as you will be using them in almost every lab for the rest of the semester!

Breadboard

A breadboard is used to prototype (build and test) electronic circuits without soldering. Wires and circuit component leads push into the holes in the board, holding the components physically and connecting them electrically. In the lab, we have the powered Proto-Board PB-503 which has built-in power supplies (see Figure 1).

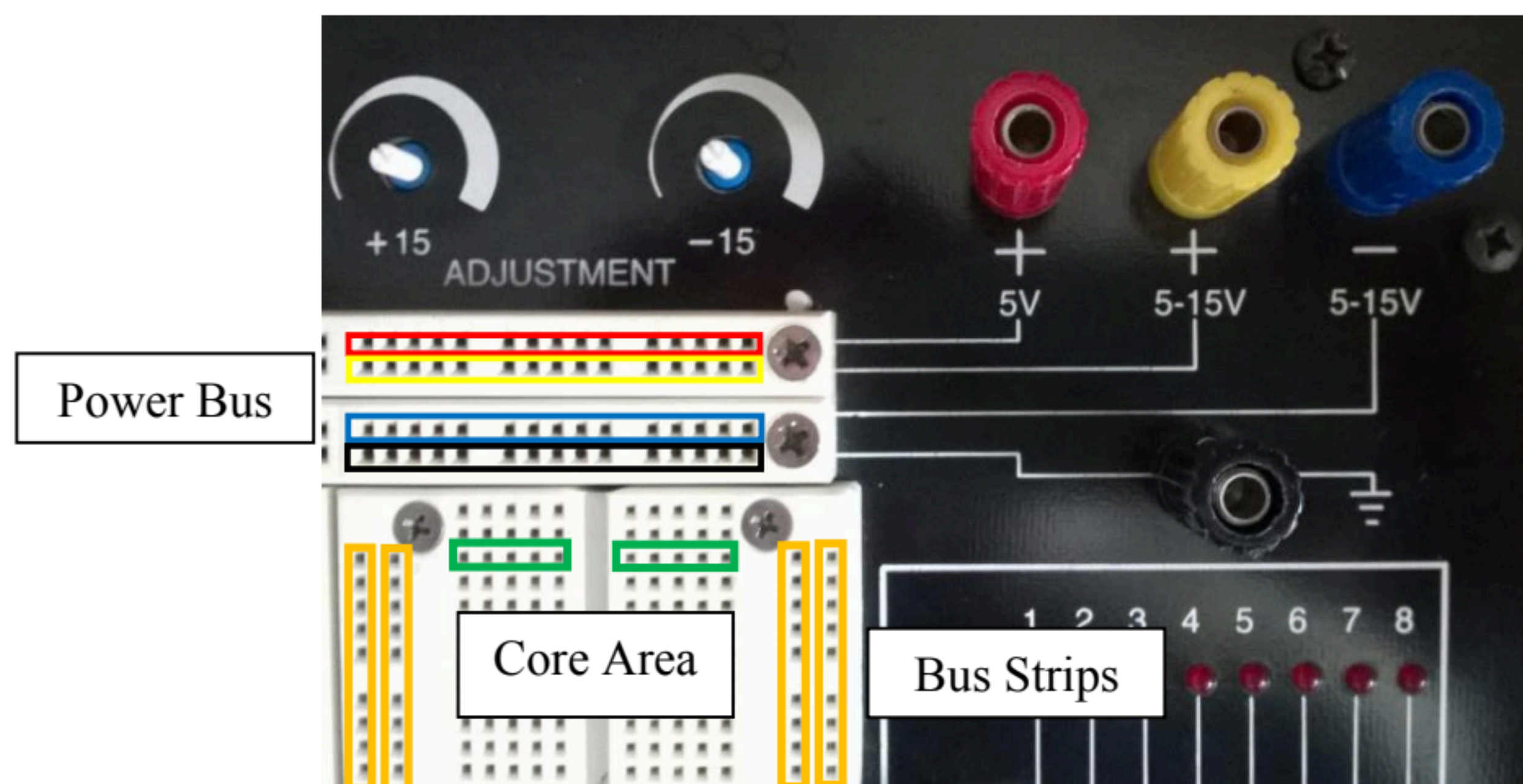


Figure 1: PB-503 breadboard used in the lab with electrical connections indicated.

Bus lines along the sides of the board are used to distribute power and reduce the tangle of wires that can develop during prototyping. In the PB-503, four additional horizontal bus lines run across the top of the breadboard. These are internally connected to the red, yellow, blue, and black terminals of the built-in power supply. The holes in a given row are interconnected as indicated in Figure 1. The two vertical bus lines along each side of a breadboard section are used to bring power down to the circuit components. Holes in a given column are interconnected, but the upper and lower half are not connected to each other in the older type boards. The vertical bus lines are not internally connected to the power supply. Use jumper wires to connect these to the horizontal power bus lines as needed.

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The core area of the breadboard is where the components are installed. The holes in a given row are connected, but there is a break in this connection across the “trough” that runs down the center of the core area.

Familiarize yourself with the breadboard layout. Figure out which pins are connected internally! If you need help, ask your TA or instructor.

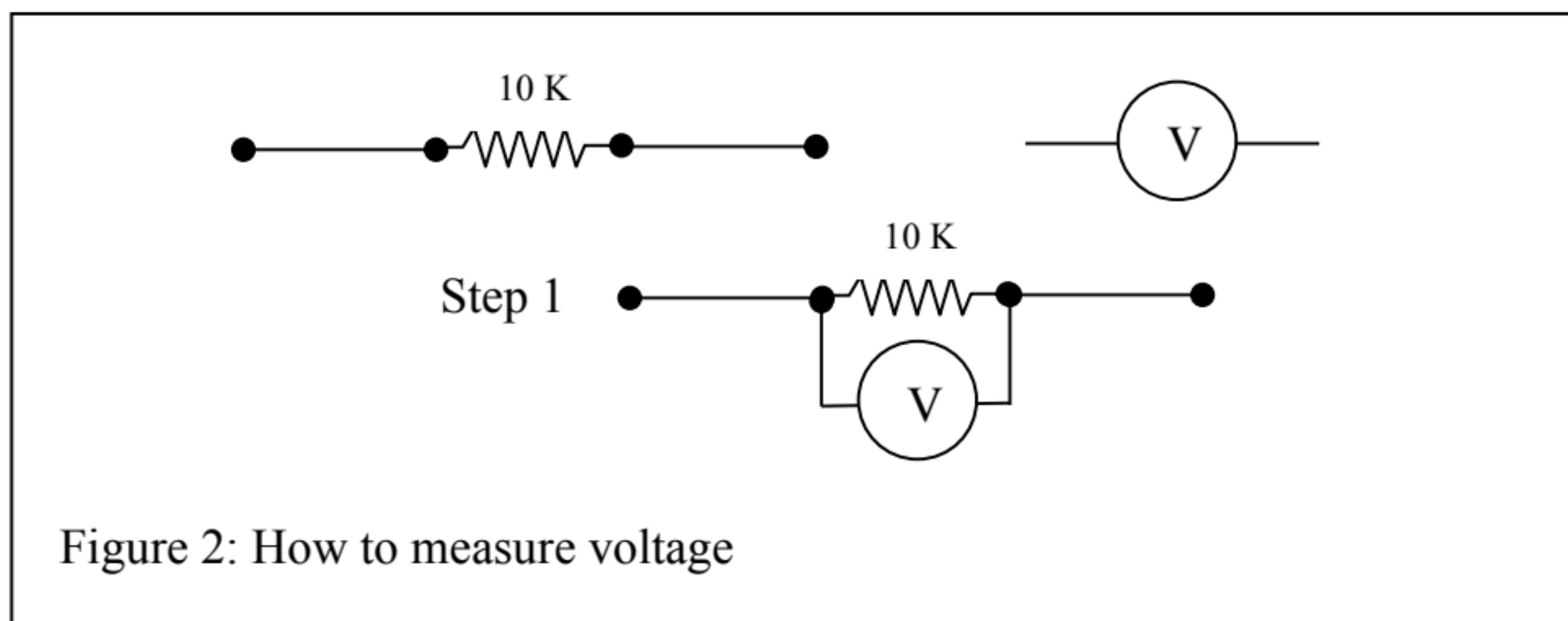
Digital Multimeter (DMM)

The digital multimeter is a versatile instrument used for measuring AC and DC voltages and currents, as well as ohmic resistance, continuity, etc. In the lab, you will use the **Agilent 34450A** (and occasionally Keithley 179A) DMM. The user manual of the device is available in the lab. For estimates of the measurement uncertainties, refer to the data sheet in the Appendix.

One important feature of this DMM is that it has 6 input terminals. It is very important to understand which pair of terminals to use for various measurements (see Figure A.1. in the Appendix). Another interesting feature that should be noted is the frequency range of the AC measurements. It is possible in some experiments to encounter frequencies outside the range of this instrument, and there is no warning that it is giving a faulty reading, except that the readings on different scales do not agree.

How to measure voltages and currents with a DMM

To measure the voltage difference between two points of a circuit, first connect the test leads to the appropriate terminals of the DMM (as shown in Figure A.1 in the Appendix), press the button for DCV (or ACV) on the DMM, and then touch the test leads to the points in the circuit (see Figure 2). A DMM used for measuring voltages usually has a large resistance ($\sim 10\text{ M}\Omega$) and since it is connected parallel has very little impact on the circuit and can be considered external to the circuit

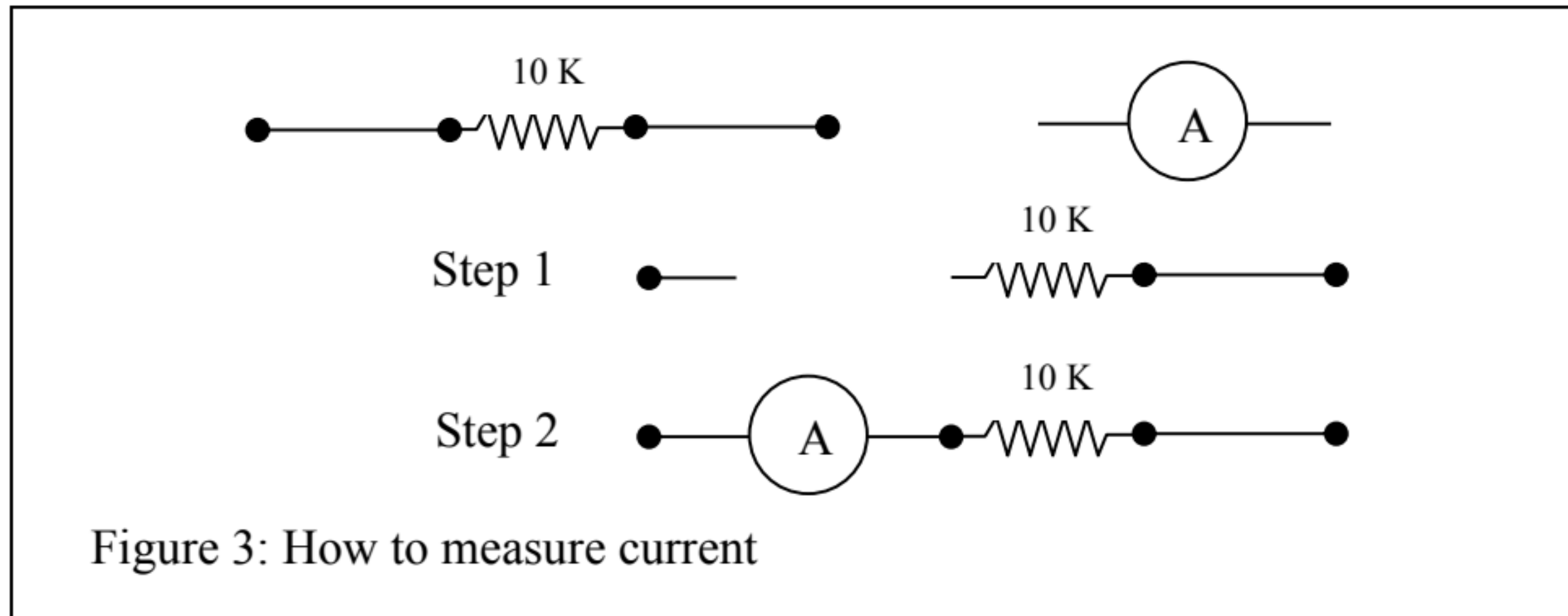


To measure current through a circuit component, you need to break the circuit and insert the DMM in the gap in series with the component (see Figure 3). A DMM measuring current must become physically part of the original circuit so the current can flow through it (its resistance therefore must be very small). First, connect the test leads to the appropriate terminals of the DMM depending on the magnitude of the current (see Figure

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A.1 in the Appendix) and then connect the leads to the circuit. Do not forget to press DCI (or ACI) on the DMM to enable the current measurement (Agilent 34450A).



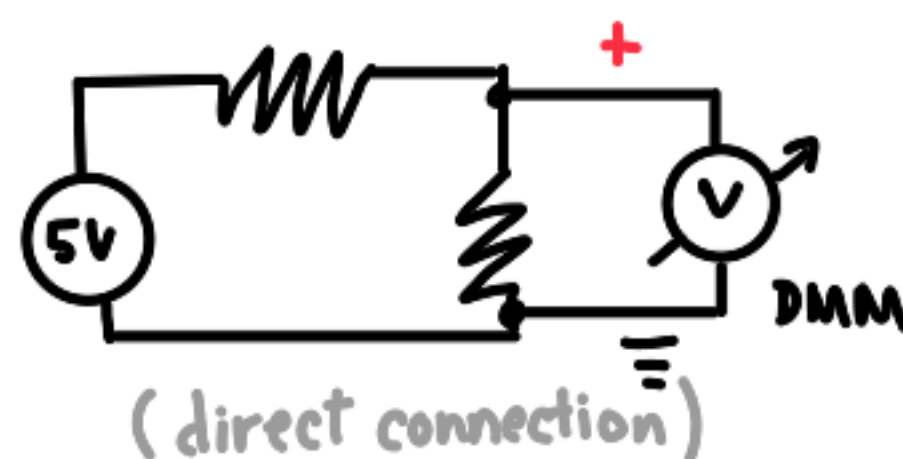
Impedance Bridge (LCR meter)

There are three different devices in the lab to measure the impedance of reactive components (capacitors and inductors): a Stanford Research Model 715, a BK Precision Model 891, and a Keysight U1733C. They can also be used for resistance measurements, but these can be done more conveniently with a DMM (note that the DMMs in the lab can also measure capacitance). They are simple to use and it should be employed to measure the actual as opposed to “nominal” values of fixed capacitors or inductors before inserting them in a circuit. The accuracy is better than 0.05% (0.2% in case of the SR 715) for the R, L, and C components commonly used in this lab.

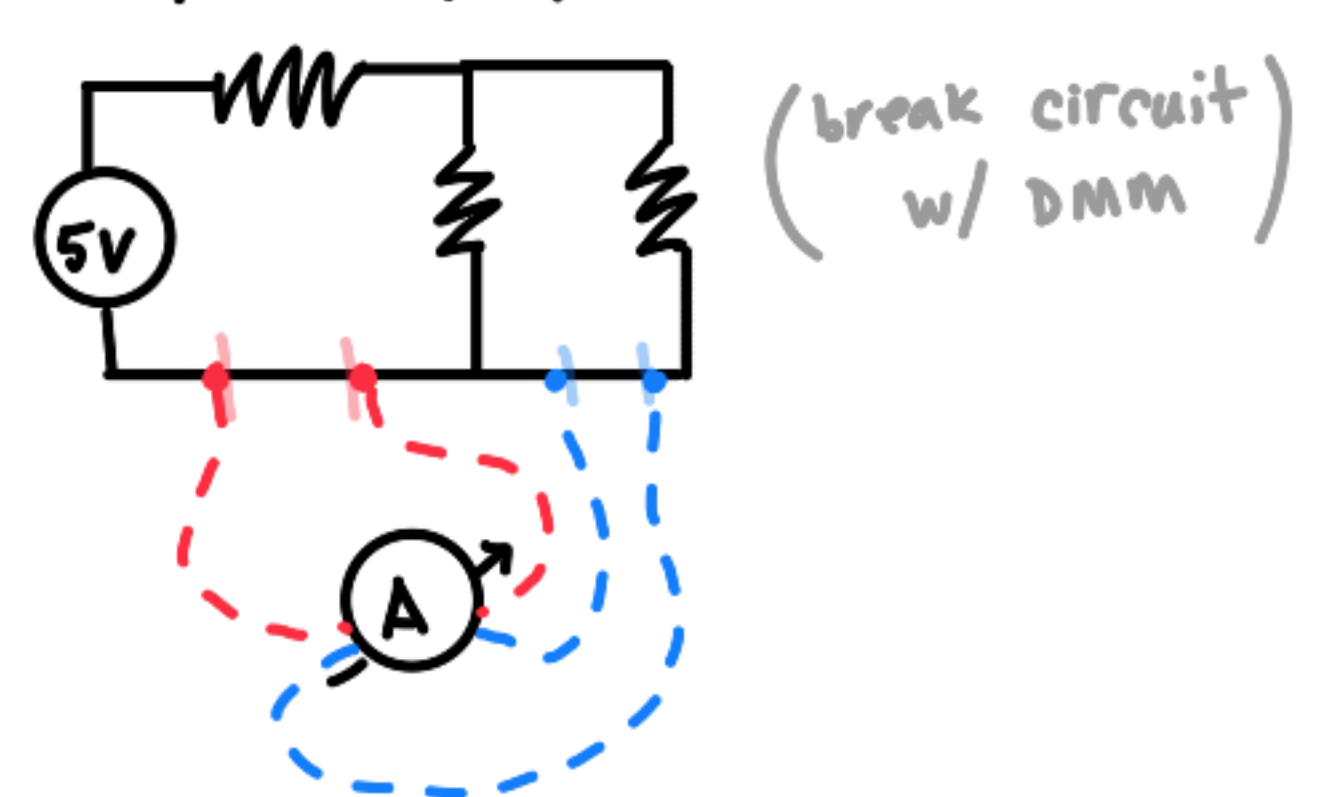
Ohm-meter (Ω)



voltmeter (V)



ampmeter (A)



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Part 1a: Voltage divider

This circuit shown on Figure 4 is very simple. It consists of two resistors in series connected to a DC power supply. The figure on the left shows the schematic diagram that you may have seen before in textbooks. The triangular symbol at the bottom represents the reference point (ground or GND) which we call 0V in the circuit. Note that this point may not be an actual 0V (i.e. Earth ground) but all other voltages are referenced relative to this point.

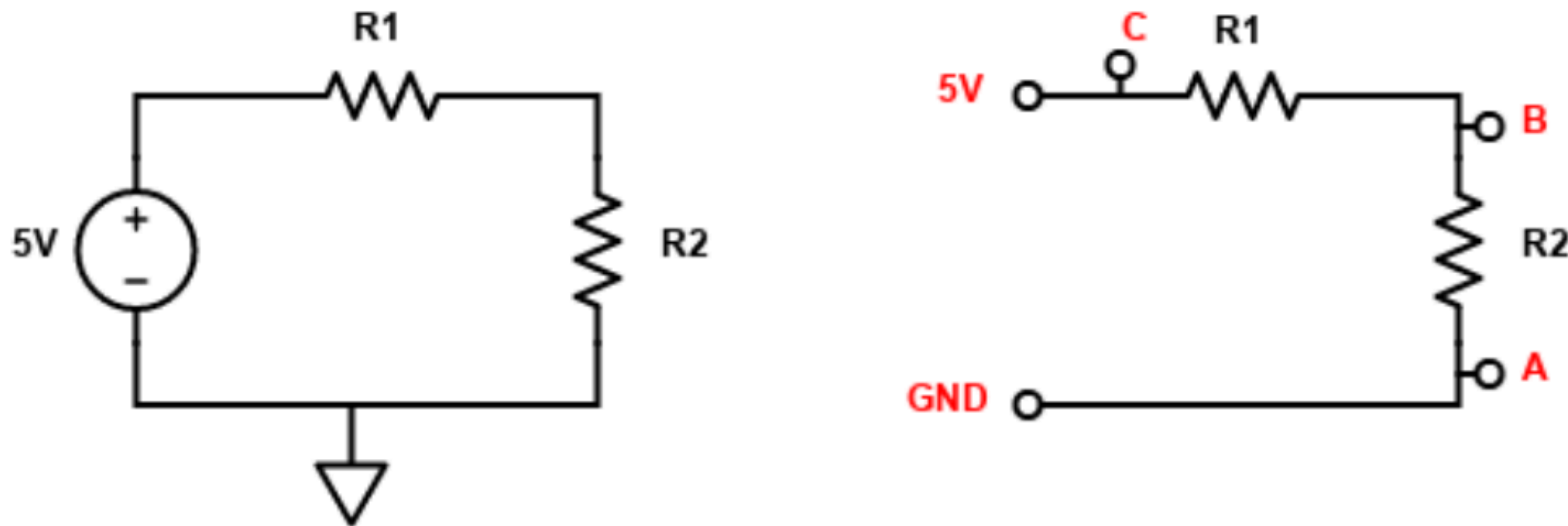


Figure 4. Voltage divider. Schematic diagram on the left and connection points to power supply and test instrument (DMM) indicated on the right.

The figure on the right shows how to build the circuit on the breadboard and connect it to the DC power supply, which can be either the built-in 5V power supply of the breadboard in the lab or an external (standalone) power supply.

Use resistors $R_1=10\text{ k}\Omega$ and $R_2=20\text{ k}\Omega$. Measure the actual values of the resistors with the DMM before you insert them in the breadboard. The actual values may differ from the nominal values by several %.

$$R_1 = \underline{9.882} \text{ [k}\Omega\text{]} \quad R_2 = \underline{19.847} \text{ [k}\Omega\text{]}$$

Measure the actual voltage of the power supply with the DMM:

$$V = \underline{5.0426} \text{ [V]}$$

Calculate the expected voltage V_{ba} (between points **B** and **A**) and V_{cb} (between points **C** and **B**).

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$$\begin{aligned}
 V_{\text{tot}} &= V_{ab} + V_{bc} \\
 &= IR_1 + IR_2 \\
 5.0426 \text{ V} &= I(R_1 + R_2) \\
 5.0426 \text{ V} &= I(9.882 + 19.847) \\
 5.0426 \text{ V} &= I(29.729) \\
 I &= \frac{5.0426 \text{ V}}{29.729 \Omega} = 1.6962 \times 10^{-4} \text{ A}
 \end{aligned}$$

$$\begin{aligned}
 V_{\text{tot}} &= (9.882 \text{ k}\Omega \cdot 1.6962 \times 10^{-4} \text{ A}) + (19.847 \text{ k}\Omega \cdot 1.6962 \times 10^{-4} \text{ A}) \\
 &= 1.6762 \text{ V} + 3.3664 \text{ V} \\
 &\quad V_{ab} \quad \quad \quad V_{bc}
 \end{aligned}$$

Expected voltages: $V_{ba} = 1.6762$ [V] $V_{cb} = 3.3664$ [V]

Measure V_{ba} and V_{cb} using the DMM.

Measured voltages: $V_{ba} = 3.3642$ [V] $V_{cb} = 1.673$ [V]

Do the two voltages agree? If the voltages do not agree within 1%, check your calculations and measurements!

Calculate the current flowing in the circuit.

$$1.6962 \times 10^{-4} \text{ A}$$

Expected current: $I = 169.62$ [mA]

Measure the current in the circuit using the DMM. Connect the + test lead to the 10A (or 100 mA) terminal of the DMM (see Figure A.1 in the Appendix) after you connected the far end of the leads to the appropriate places in the circuit!

Measured current: $I = 169.11$ [mA]

Do the currents agree (within 1%)? If they do not agree, repeat your calculations and measurements!

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As an additional check, use **Ohm's Law** to calculate the voltage V_{ba} . Use the actual value of the R_2 resistor and the measured current I . Does it agree with the measured V_{ba} (within 1%)?

$$V_{ba} = 169.11 \text{E-6} \cdot 19.847 \text{E-3} = 3.35 \text{ V } \checkmark$$

Answer: $V_{ba} = IR = \underline{3.35} \text{ [V]}$

Calculate the total electrical power P_{total} that flows from the power supply into the circuit:

$$P = \frac{V^2}{R} = VI = I^2 R$$

$$\hookrightarrow 5.0426 (1.696 \text{E-4}) = 8.55 \text{E-4 W}$$

Answer: $P_{total} = \underline{0.855} \text{ [mW]}$

What percentage of the total power is dissipated in the 20 K resistor?

$$P_{R2} = I^2 R$$

$$\hookrightarrow (1.696 \text{E-4})^2 \cdot (19847)$$

$$\Rightarrow 5.71 \text{E-4 W} = 0.571 \text{ mW}$$

$$\Rightarrow \% P = \frac{0.571}{0.855} \cdot 100\% = 66.8\%$$

Answer: $\underline{66.8} \text{ \% of the total power}$

Question:

Resistors have a maximum power rating (1/4 Watt to 1 Watt for most of the resistors in the lab). If the maximum ratings are exceeded the resistors may overheat ("fry"). As an exercise, calculate the maximum "safe" current I_{max} through a 2 k Ω resistor rated at 1/4 Watt and the maximum "safe" voltage V_{max} across that same resistor.

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$$P = \frac{V^2}{R} = I^2 R : 0,25W = \frac{V_{max}^2}{2k\Omega} = I_{max}^2 \cdot 2k\Omega$$

$$\hookrightarrow V_{max} = \sqrt{500} \\ = 22,36V$$

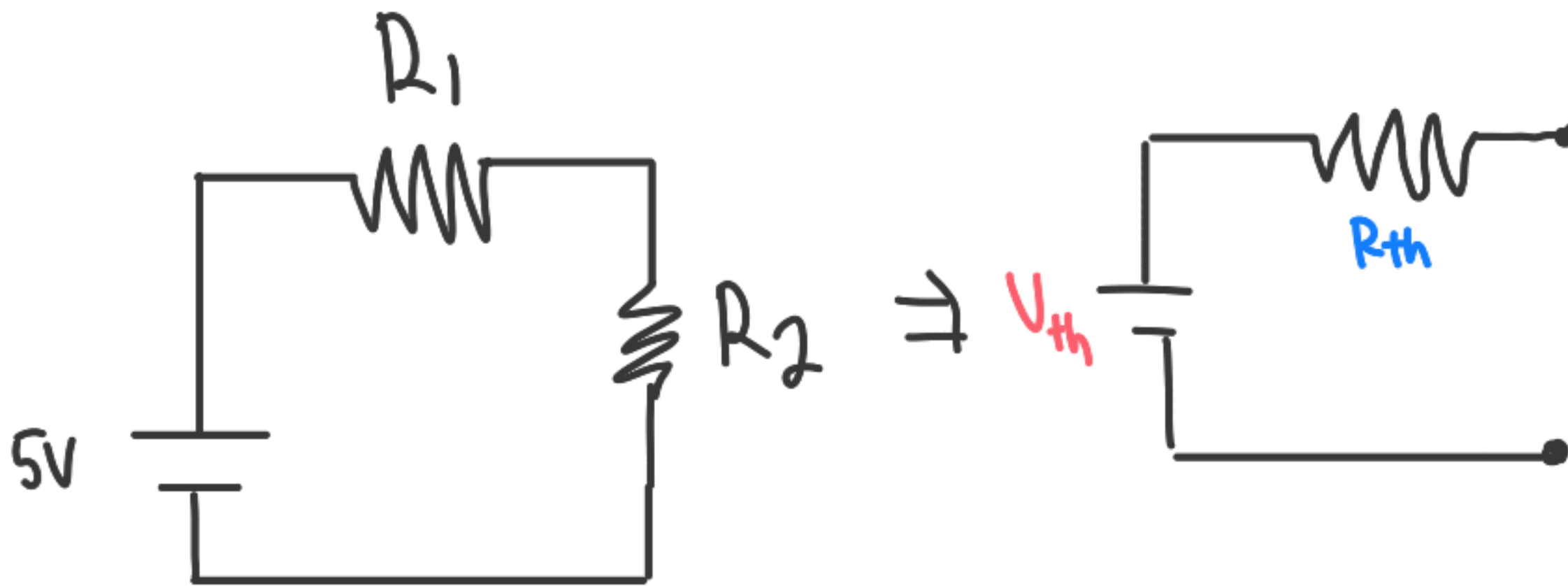
$$\hookrightarrow I_{max} = \sqrt{\frac{0,25}{2000}} \\ = 11,18mA$$

Answers:

$$I_{max} = \underline{0.0118} [A] \quad V_{max} = \underline{0.325} [V]$$

Part 1b: Thevenin equivalent circuit

Draw the Thevenin equivalent circuit for the voltage divider used in Part 1a with the output taken across the R_2 resistor (V_{ba}). Determine $V_{Thevenin}$ and $R_{Thevenin}$.



$$V_{th} = v\left(\frac{R_{ba}}{R_{eq.}}\right) = 5\left(\frac{19.847}{29.729}\right) = 5 \cdot 0.66759 = 3.3379V$$

$$R_{th} = 9.882k\Omega$$

Answers:

$$V_{Th} = \underline{3.3379} [V] \quad R_{Th} = \underline{9.882E3} [\Omega]$$

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Part 2a: Adding a load to the circuit changes the output voltage.

We can view the voltage divider used above as a circuit with an input voltage (5V from the power supply) and an output voltage taken across the R_2 resistor (V_{ba}). You can see that by changing R_2 you can adjust the output voltage to any value between 0V and the maximum voltage of the power supply (5V in this case). The following exercise demonstrates what happens to the output voltage of a circuit if you connect a load to its output.

Modify the circuit as shown in Figure 5 by adding a load resistor (R_L) to your earlier setup. Try two different load resistors, a large one ($R_L \gg R_2$) and a small one ($R_L \lesssim R_2$). Note: you should get in the habit of measuring all relevant resistor values and use the measured values in calculations!

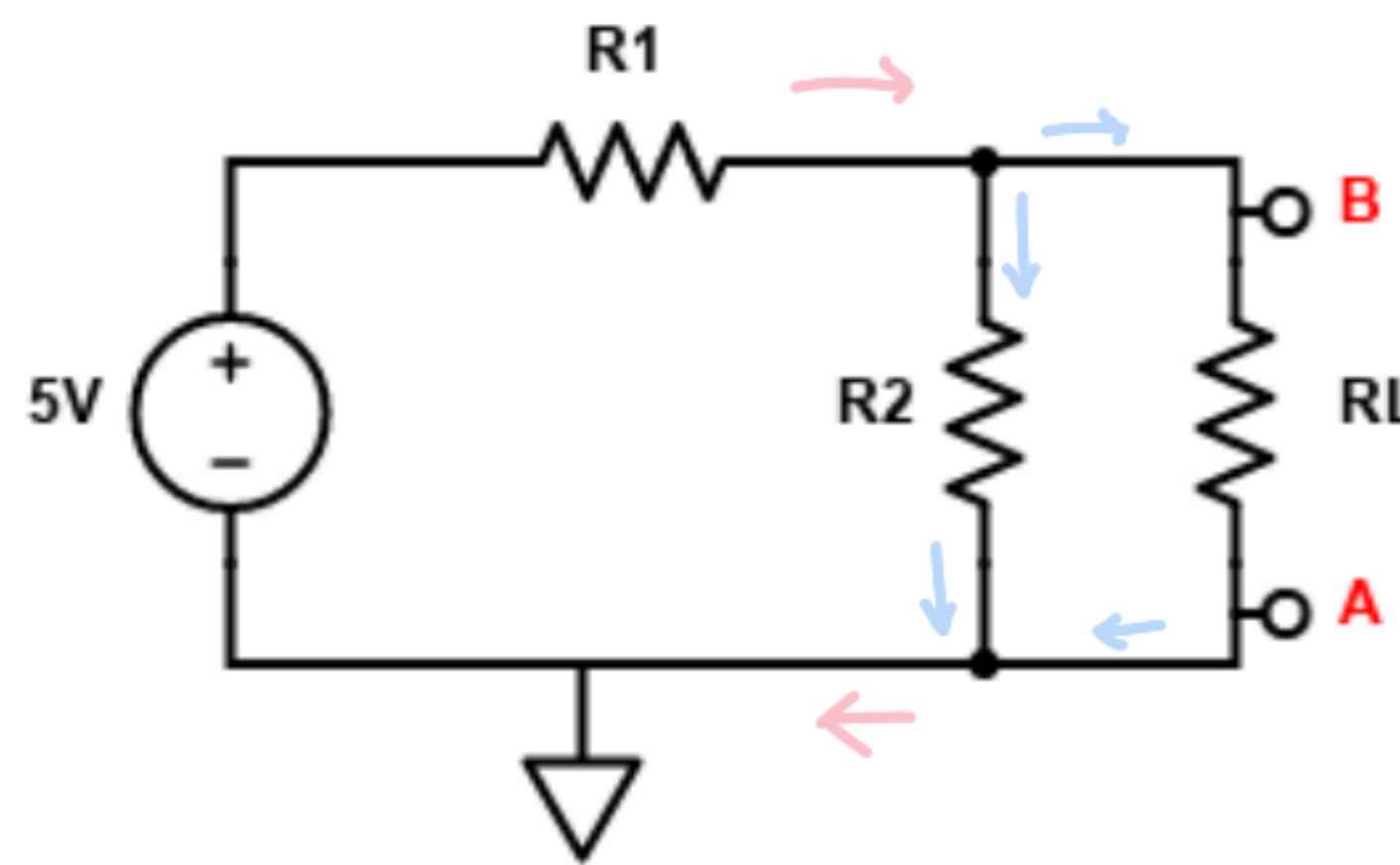


Figure 5. Voltage divider with load.

Start with a load resistor $R_L = 1 \text{ M}\Omega$. Measure its actual resistance using the DMM:

$$R_L = \underline{994} \text{ [k}\Omega\text{]}$$

Calculate the expected voltage V_{ba} of point B with respect to point A. Hint: This calculation is easier if you use the Thevenin equivalent circuit.

$$R_{eq} = \frac{994 \text{ k}\Omega \cdot 19.847}{994 \text{ k}\Omega + 19.847} = 19.458 \text{ k}\Omega$$

$$V_{th} = 5V \cdot \frac{R_{eq}}{R_1 + R_{eq}} = 5.043 \left(\frac{19.458}{9.882 + 19.458} \right) = 3.34$$

$$\text{Expected voltage: } V_{ba} = \underline{3.34} \text{ [V]}$$

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Measure V_{ab} using the DMM.

Measured voltage: $V_{ba} = \underline{3.342}$ [V]

Change the load resistor to $R_L = 20 \text{ k}\Omega$ and repeat the exercise. Measure its actual resistance using the DMM:

$R_L = \underline{19.835}$ [k Ω]

Calculate the expected voltage V_{ba} of point **B** with respect to point **A** (again this is easier using the Thevenin equivalent circuit).

$$R_{eq} = \frac{19.835 \text{ k}\Omega \cdot 19.847 \text{ k}\Omega}{19.835 \text{ k}\Omega + 19.847 \text{ k}\Omega} = 9.92 \text{ k}\Omega$$

$$V_{th} = \left(\frac{9.920 \text{ k}\Omega}{9.982 \text{ k}\Omega + 9.92 \text{ k}\Omega} \right) 5.0426 \text{ V}$$

Expected voltage: $V_{ba} = \underline{2.53}$ [V]

Measure V_{ab} using the DMM.

Measured voltage: $V_{ba} = \underline{2.47}$ [V]

Conclusion: How does the load affect the output voltage (V_{ba}) with respect to the case when there was no load connected to the voltage divider output in Part 1?

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Part 2b: Measuring voltage across a large resistor may give surprising result!

Remove the load resistor and set up the voltage divider in Figure 4 using $R_1=R_2=10\text{ M}\Omega$ resistors.

Calculate the expected voltage V_{ba} of point **B** with respect to point **A**.

Expected voltage: $V_{ba} = \underline{2.5}$ [V]

Measure V_{ba} using the DMM.

Measured voltage: $V_{ba} = \underline{1.66}$ [V]

Do the two voltages agree? The measured voltage will be smaller than the calculated voltage. Why? Hint: Check the input impedance of the DMM in the Appendix. How does it affect the voltage measurement?

no, the two voltages appear not to agree at first glance, however, this is due to the impedance of the DMM- which impedes the flow of electrons.

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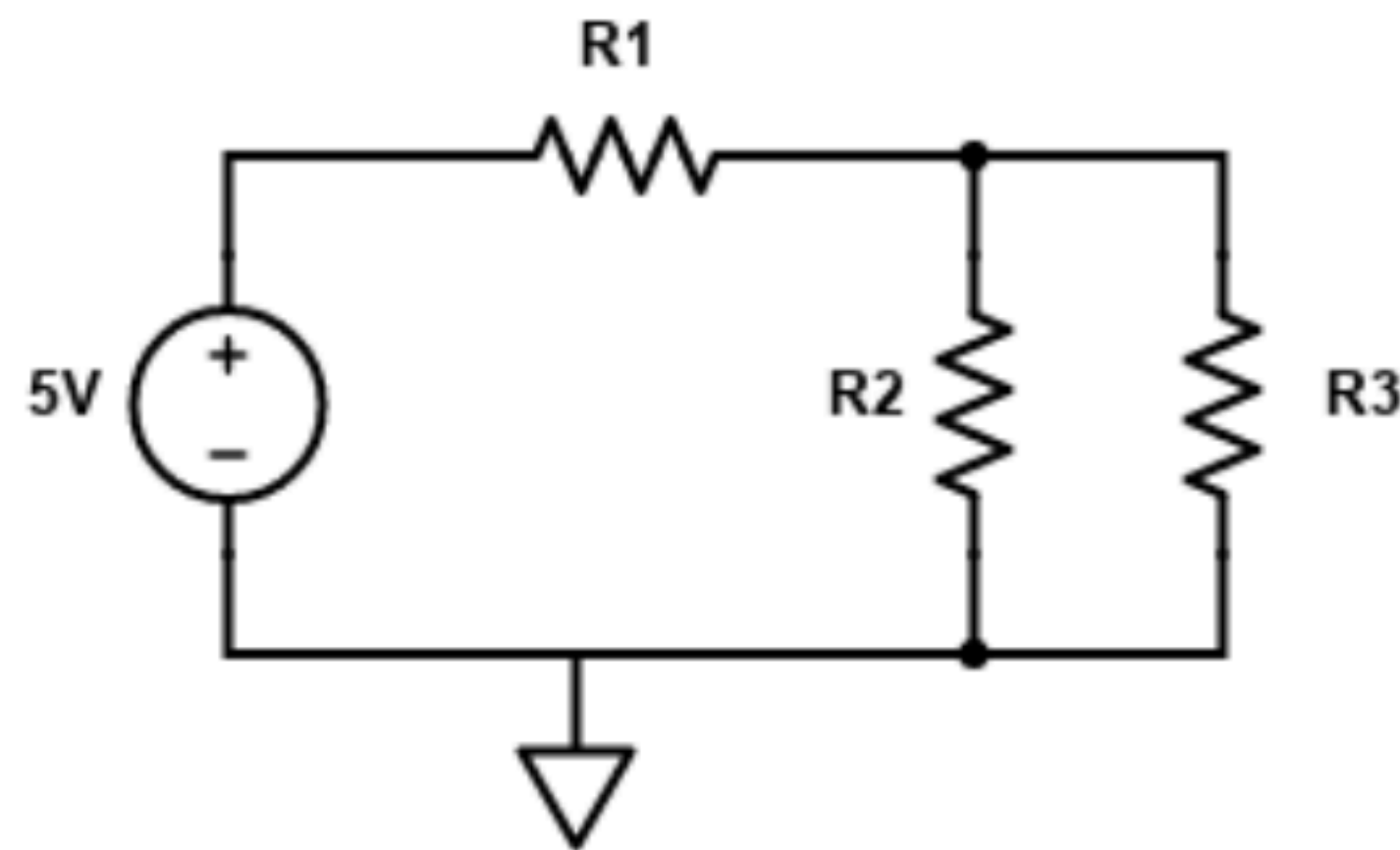
Part 3: Current dividerBuild the circuit shown in Figure 6 using $R_1 = 10 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $R_3 = 4.7 \text{ k}\Omega$.

Figure 6. Current divider.

Measure the actual resistances using the DMM:

$$R_1 = \underline{9.882} \text{ [k}\Omega\text{]} \quad R_2 = \underline{0.987} \text{ [k}\Omega\text{]} \quad R_3 = \underline{4.676} \text{ [k}\Omega\text{]}$$

Measure the actual voltage of the power supply using the DMM:

$$V = \underline{4.845} \text{ [V]}$$

Calculate the current through each resistor (attach on a separate sheet if you need more space).

$$I_1 = \frac{V}{R_T} \Rightarrow R_T = R_1 + \overset{R_{11}}{\frac{R_2 \cdot R_3}{R_2 + R_3}} = \frac{0.987 \cdot 4.676}{0.987 + 4.676} = 0.8149 = R_{11}$$

$$= \frac{4.845}{10.6969} = 0.4529 \text{ mA} \quad \hookrightarrow R_T = R_{11} + 9.882 \text{ k}\Omega = 10.6969 \text{ k}\Omega$$

$$V_{23} = I_1 R_{11} = 0.4529 \text{ mA} \cdot 0.8149 \text{ k}\Omega = 0.369 \text{ V}$$

$$I_2 = \frac{V_{23}}{R_2} = \frac{0.369 \text{ V}}{0.987 \text{ k}\Omega} = 0.374 \text{ mA}$$

$$I_3 = \frac{V_{23}}{R_3} = \frac{0.369 \text{ V}}{4.676 \text{ k}\Omega} = 0.078 \text{ mA}$$

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Expected current through R_1 : $I_1 =$ 0.4529 [mA]Expected current through R_2 : $I_2 =$ 0.374 [mA]Expected current through R_3 : $I_3 =$ 0.078 [mA]**Sum** of currents through R_2 and R_3 : $I_2 + I_3 =$ 0.452 :D [mA]If you did the calculation correctly, you should find $I_2 + I_3 = I_1$!

Now **measure** the currents. Connect the + test lead to the 10A (or 100 mA) terminal of the DMM (see Figure A.1 in the Appendix) after you connect the far end of the leads to the appropriate points in the circuit!

Measured current through R_1 : $I_1 =$ 0.453 [mA]Measured current through R_2 : $I_2 =$ 0.381 [mA]Measured current through R_3 : $I_3 =$ 0.080 [mA]

Do measured and calculated values agree? If not, check your work!

yeah!

Add-on questions:

How much power flows from the power supply into the circuit?

$$\begin{aligned}
 P &= I^2 R = \frac{V^2}{R} \\
 &= \frac{V^2}{R_T} = \frac{(4.845\text{V})^2}{10.6969\text{k}\Omega} = 2.194\text{mW}
 \end{aligned}$$

 $P_{total} =$ 2.194 [mW]

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How much electrical power is dissipated in each resistor?

$$P_1 = 0.4529^2 \cdot 9.882 = 2.026 \text{ mW}$$

$$P_2 = 0.374^2 \cdot 0.987 = 0.1381 \text{ mW}$$

$$P_3 = 0.078^2 \cdot 4.676 = 0.0284 \text{ mW}$$

Power dissipated in R_1 : $P_1 =$ 2.026 [mW]

Power dissipated in R_2 : $P_2 =$ 0.1381 [mW]

Power dissipated in R_3 : $P_3 =$ 0.0284 [mW]

If you did everything right, the total power should come out to be the sum of the powers dissipated by the resistors. If not, check your calculations.

Done

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Appendix

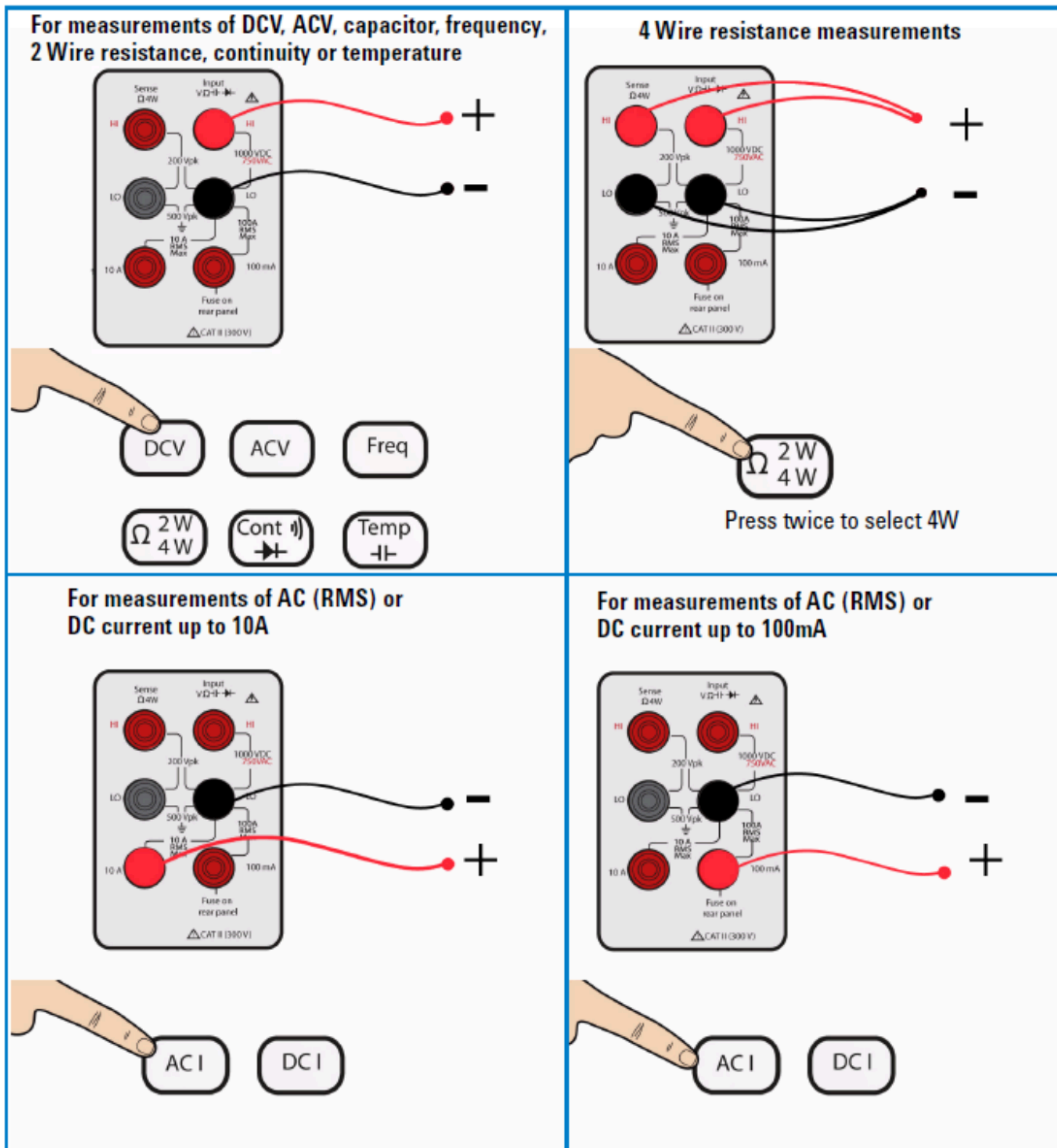


Figure A.1. Connecting test leads to the terminals of the Agilent 34450A DMM for different measurements.

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Table A.1: DC and AC specifications for Agilent 34450A DMM

DC specification

Specifications are for 90 minutes warm-up time, slow mode, and calibration temperature within 18 °C - 28 °C

Table 1. DC accuracy ± (% of reading + % of range)

Function	Range ¹	Test current or burden voltage	Input impedance	1 year 23 °C ± 5 °C	Temperature coefficient 0 °C - 18 °C 28 °C - 55 °C
DC voltage	100.000 mV	–	10 MΩ or > 10 GΩ	0.018 + 0.008	0.0020 + 0.0008
	1.00000 V	–	10 MΩ or > 10 GΩ	0.015 + 0.005	0.0015 + 0.0008
	10.0000 V	–	10 MΩ	0.015 + 0.005	0.0020 + 0.0008
	100.000 V	–	10 MΩ	0.015 + 0.005	0.0020 + 0.0008
	1000.00 V	–	10 MΩ	0.015 + 0.005	0.0020 + 0.0008
Resistance (2-wire) ²	100.000 Ω	1 mA	–	0.050 + 0.008	0.0060 + 0.0008
	1.00000 kΩ	500 μA	–	0.050 + 0.008	0.0060 + 0.0005
Resistance (4-wire) ²	10.0000 kΩ	100 μA	–	0.050 + 0.005	0.0060 + 0.0005
	100.000 kΩ	10 μA	–	0.050 + 0.005	0.0060 + 0.0005
	1.00000 MΩ	1 μA	–	0.060 + 0.005	0.0060 + 0.0005
	10.0000 MΩ	100 nA	–	0.250 + 0.005	0.0250 + 0.0005
	100.000 MΩ	100 nA/10 MΩ	–	2.000 + 0.005	0.3000 + 0.0005
DC current	100.000 μA	< 0.02 V	–	0.05 + 0.015	0.007 + 0.0015
	1.00000 mA	< 0.2 V	–	0.05 + 0.007	0.007 + 0.0010
	10.0000 mA	< 0.02 V	–	0.05 + 0.015	0.008 + 0.0015
	100.000 mA	< 0.2 V	–	0.05 + 0.007	0.008 + 0.0010
	1.00000 A	< 0.2 V	–	0.10 + 0.015	0.012 + 0.0015
	10.0000 A	< 0.6 V	–	0.25 + 0.007	0.015 + 0.0010
Continuity ³	1000 Ω	0.5 mA	–	0.05 + 0.03	0.005 + 0.005
Diode test ⁴	1.0000 V	0.5 mA	–	0.05 + 0.03	0.005 + 0.005

Notes on specifications:

- 20% over range on all ranges except 1000 VDC and 10 A range.
- Specifications are for 4-wire Ω or 2-wire Ω using NULL function. If without NULL function, add 0.2 Ω additional error.
- Continuity thresholds is fixed at less than 10 Ω
- Specifications are for the voltage measured at the input terminals only.

AC specifications

Specifications are for 90 minutes warm-up time, slow mode, and calibration temperature within 18 °C - 28 °C

Table 2. AC Accuracy ± (% of reading + % of range)

Function	Range ¹	Frequency	1 year 23 °C ± 5 °C	Temperature coefficient 0 °C - 18 °C 28 °C - 55 °C
True RMS AC voltage ²	100.000 mV	20 Hz – 45 Hz	1.0 + 0.1	0.02 + 0.02
		45 Hz – 10 kHz	0.2 + 0.1	0.02 + 0.02
		10 kHz – 30 kHz	1.5 + 0.3	0.05 + 0.02
		30 kHz – 100 kHz ³	3.0 + 0.3	0.10 + 0.02
	1.00000 V to 750.00 V	20 Hz – 45 Hz	1.0 + 0.1 ⁴	0.02 + 0.02
		45 Hz – 10 kHz	0.2 + 0.1	0.02 + 0.02
		10 kHz – 30 kHz	1.5 + 0.3	0.05 + 0.02
		30 kHz – 100 kHz ³	3.0 + 0.3 ⁵	0.10 + 0.02
True RMS AC current ²	10.0000 mA to 10.0000 A	20 Hz – 45 Hz	1.5 + 0.1	0.02 + 0.02
		45 Hz – 1 kHz	0.5 + 0.1	0.02 + 0.02
		1 kHz – 10 kHz ⁶	2.0 + 0.2	0.02 + 0.02

Notes on specifications:

- 20% over range on all ranges except ACV 750 V and ACI 10 A
- Specifications are for sine-wave inputs more than the 5 % of range except 750 V range. Input signal must be more than 50 Vrms for 750 V range. Maximum crest factor of 3 at full scale. Input impedance is 1 MΩ in parallel with capacitance less than 120 pF, AC couple with up to 400 DCV
- Additional error to be added as frequency > 30 kHz and signal input < 10% of range. 30 kHz to 100 kHz: 0.003% of full scale per kHz
- For input < 200 V rms
- For input < 300 V rms
- Frequencies > 5 kHz are typical for all ranges

Table A.1 gives the uncertainty as ±(% of reading + % of range).

For example, a reading of 1.5 DCV obtained in the 10V range has an uncertainty of
 $\pm[(0.00015)(1.5 \text{ V}) + (0.00005)(10 \text{ V})] = \pm[0.225\text{mV} + 0.5 \text{ mV}] = \pm0.725 \text{ mV}$

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Table A.2: Spec sheet for the Keithley Model 179A Digital Multimeter

	Range	Resolution	Uncertainty	
Volts			DC	TRMS AC (100 Hz – 10 kHz)
	200 mV	10 μ V	0.04% + 3d	0.7% + 15d
	2 V	100 μ V	0.04% + 1d	0.6% + 15d
	20 V	1 mV	0.04% + 1d	0.5% + 15d
	200 V	10 mV	0.04% + 1d	0.5% + 15d
	1000 V	100 mV	0.04% + 1d	0.5% + 15d
Amps			DC	TRMS AC (45 Hz – 10 kHz)
	200 μ A	10 nA	0.2% + 2d	1% + 15d
	2 mA	100 nA	0.2% + 2d	1% + 15d
	20 mA	1 μ A	0.2% + 2d	1% + 15d
	200 mA	10 μ A	0.2% + 2d	1% + 15d
	2000 mA	100 μ A	0.2% + 2d	1% + 15d
	20 A	1 mA	0.5% + 2d	1% + 15d
Ohms			HI (2 V Max)*	LO (0.2 V Max)*
	2 k Ω	100 m Ω		0.15% + 15d
	20 k Ω	1 Ω	0.04% + 1d	0.15% + 15d
	200 k Ω	10 Ω	0.04% + 1d	0.15% + 15d
	2000 k Ω	100 Ω	0.04% + 1d	0.15% + 15d
	20 M Ω	1 k Ω	0.10% + 1d	0.15% + 15d

*The nominal applied current for each range = V_{max}/R_{max} Table A.2 gives the uncertainty as $\pm$ (% of reading + number of digits).

The number of digits is given in terms of the resolution of the range used to make the measurement: 1d is equivalent to the value of 1 in the last digit of the display, which is identical to the resolution; 2d is equivalent to the value of 2 in the last digit of the display, which is identical to twice the resolution...

For example, a reading of 1.5 DCV obtained using the 2 V range has an uncertainty of

$$\pm[(0.0004)(1.5 \text{ V}) + (1)(100 \mu\text{V})] = \pm[600 \mu\text{V} + 100 \mu\text{V}] = \pm 700 \mu\text{V}$$

This same reading taken using the 20 V range would have an uncertainty of

$$\pm[(0.0004)(1.5 \text{ V}) + (1)(1 \text{ mV})] = \pm[0.6 \text{ mV} + 1 \text{ mV}] = \pm 1.6 \text{ mV}$$

Resolution has a significant effect on uncertainty. Always use the lowest range possible when making a measurement.